DSA ki Baithak

CODE BAITHAK

BY NAMAN BHAIYA

Teaser / Trailer :

```
*   *   *
  *   *
*   *
  *
*
```

```
* * * * *
  * * * *
    * * *
      * *
        *
      * *
    * * *
  * * * *
* * * * *
```

```
    *
  *   *
*       *
*       *
  *   *
    *
```

```
4           4
4 3       3 4
4 3 2     2 3 4
4 3 2 1   1 2 3 4
4 3 2 1 0 1 2 3 4
4 3 2 1   1 2 3 4
4 3 2     2 3 4
4 3       3 4
4           4
```

```
        1
      2 1   1 2
    3 2 1     1 2 3
4 3 2 1         1 2 3 4
    3 2 1     1 2 3
      2 1   1 2
        1
```

```
5 5 5 5 5 5 5 5 5
5 4 4 4 4 4 4 4 5
5 4 3 3 3 3 3 4 5
5 4 3 2 2 2 3 4 5
5 4 3 2 1 2 3 4 5
5 4 3 2 2 2 3 4 5
5 4 3 3 3 3 3 4 5
5 4 4 4 4 4 4 4 5
5 5 5 5 5 5 5 5 5
```

"Patterns Ka Gyaan...."

* Patterns always get printed rowwise (from left to right)

* Outer loop runs for the no. of rows

  &

  Inner loop runs for the no. of columns

Patterns Halwa Recipe

Steps -

1. First find out the total no. of rows in the pattern.
2. Find the tasks to be performed in each row of pattern.
3. Perform all the tasks of the current row.
4. Do preparation of the next row & change the line.

Home Work - Question

20.

$n = 5$

```
    - - - - 1
    - - - 1 2 1
    - - 1 2 3 2 1
      1 2 3 4 3 2 1
0   1 2 3 4 5 4 3 2 1
    - 1 2 3 4 3 2 1
    - - 1 2 3 2 1
          1 2 1
            1
```

① Rows → $2*n-1$

② Space, Digits → row

Space	Digits
4	1
3	3
2	5

-1 → +1

+2 → -2

③ ✓

21.

$n = 5$

```
* * * * * * * * *
* * * * * * * * *
* * * * * * * * *
* * * * * * * *
* * * * * * * * *
```

① Rows → n

② Star, Space, Star → row

Star	Space	Star
5	0	-1
4	1	
3	3	
2	5	
	7	

-1 +2

③ ✓

④ —

22.

col → 2*n-1-2

space → 2*n-3

n = 5

```
*  -  -  -  -  -  -  -  *
*  *  -  -  -  -  -  *  *
*  *  *  -  -  -  *  *  *
*  *  *  *     *  *  *  *
*  *  *  *  *  *  *  *  *
*  *  *  *     *  *  *  *
*  *  *        *  *  *
*  *              *  *
*                    *
```

① Rows → 2*n-1

② Star, space, Star → row

Star	space
1	7
+1 2	5 -2
3	3

④ ✓

23.

$n = 7$

$n/2 \rightarrow 7/2 \rightarrow 3$

```
1 * * *   * * *
2 * *     * *
3 *       *
4
5 *       *
  * *     * *
  * * *   * * *
```

① Rows $\rightarrow$ n

② Star, space, Star $\rightarrow$ row

Star	space	Star
3	1	
2	3	
1	5	
0	7	
1	5	

-1 +2 -2

+1

③ ✓

④ —

24.

$n = 7$

```
- - - * -
- - * - *
- * - - - *
* - - - - - *
- * - - - *
- - * - *
- - - * -
```

① Rows → n

② Spaces, 1-star, spaces, 1 star

L R

$n/2$ ← 3 — 0 (−1) — row

2 — 1

(−1) 1 — 3 (+2) — 1

0 — 5 — –

(+1)

(−2)

② ✓

④

25.

$n = 9$ $\quad 9/2 \rightarrow 4$

```
* * * * * * * * *
_ * _ _ _ _ _ *
_ _ * _ _ _ *
      *   *
        *
```

① Row $\rightarrow n/2$

② Left space, *, Right spaces, * $\rightarrow$ row

Left space $\downarrow$ 1, 2

Right spaces: 5 $(n/2+1)$, 3

+1 $\quad$ −2

③

④ _____

26.

Col → (2*n + 1) − 2

Space →

n = 4

```
4 - - - - - - - 4
4 3 - - - - - 3 4
4 3 2 - - - 2 3 4
4 3 2 1   1 2 3 4
4 3 2 1 0 1 2 3 4
4 3 2 1 - 1 2 3 4
4 3 2 - - - 2 3 4
4 3           3 4
4             4
```

① Rows → 2*n+1

② digits, space, digits → row

digits → 1, 2, 3, 4 (+1)

space → 7, 5, 3 (−2)

digits → 2n−1 → 2*4−1 = 7

−1

+2

③ ____

④ ____

27.

$n = 7$

```
- - - - - - 1
- - - - 2 1 - 1 2
- - 3 2 1 - - - 1 2 3
4 3 2 1 - - - - - 1 2 3 4
- 3 2 1         1 2 3
- - - - 2 1   1 2
            1
```

① Rows → n

② left space, digits, Right space, digits → row

left space	digits	Right space
6	1	~~0~~ -1
4 -2	2 +1	1
2	3	3 +2
		5

③ _____

④ _____

28. $n = 5$

0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	0
0	1	2	2	2	2	2	1	0
0	1	2	3	3	3	2	1	0
0	1	2	3	4	3	2	1	0
0	1	2	3	3	3	2	1	0
0	1	2	2	2	2	2	1	0
0	1	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0

Home Work - Question